

Magnetic Force Generation via Hexapole Electromagnetic Actuators: Hall-Sensor-Based Modeling and Calibration Approach:

Magnetic Force Generation:

$$\mathbf{m}(\mathbf{p}) = \underbrace{\frac{3V}{\mu_o} \left(\frac{\mu - \mu_o}{\mu + 2\mu_o} \right)}_{M g_B} \mathbf{b}(\mathbf{p})$$

$$\mathbf{f}(\mathbf{p}) = \frac{1}{2} \nabla(\mathbf{m} \cdot \mathbf{B}) = \frac{3V}{2\mu_o} \left(\frac{\mu - \mu_o}{\mu + 2\mu_o} \right) \nabla(\mathbf{B} \cdot \mathbf{B})$$

Position-dependent $6 \times 6 \times 3$ dimensionless spatial-distribution tensor $\mathbf{L}(\bar{\mathbf{p}})$:

$$\bar{\mathbf{p}} = \mathbf{p}/\ell$$

$$\begin{cases} \bar{\mathbf{r}}_n = \bar{\mathbf{p}} - \bar{\mathbf{p}}_{c_n} & ; \quad \bar{r}_n = \|\bar{\mathbf{r}}_n\| \\ \bar{\mathbf{r}}_m = \bar{\mathbf{p}} - \bar{\mathbf{p}}_{c_m} & ; \quad \bar{r}_m = \|\bar{\mathbf{r}}_m\| \end{cases}$$

$$\begin{aligned} \nabla \left(\frac{\mathbf{r}_n \cdot \mathbf{r}_m}{r_n^3 r_m^3} \right) &= \frac{1}{r_n^3 r_m^3} \left[\left(1 - \frac{3(\mathbf{r}_n \cdot \mathbf{r}_m)}{r_n^2} \right) \mathbf{r}_n + \left(1 - \frac{3(\mathbf{r}_n \cdot \mathbf{r}_m)}{r_m^2} \right) \mathbf{r}_m \right] \\ &= \frac{1}{\ell^5} \frac{1}{\bar{r}_n^3 \bar{r}_m^3} \underbrace{\left[\left(1 - \frac{3(\bar{\mathbf{r}}_n \cdot \bar{\mathbf{r}}_m)}{\bar{r}_n^2} \right) \bar{\mathbf{r}}_n + \left(1 - \frac{3(\bar{\mathbf{r}}_n \cdot \bar{\mathbf{r}}_m)}{\bar{r}_m^2} \right) \bar{\mathbf{r}}_m \right]}_{\mathbf{L}(\bar{\mathbf{p}}; n, m, :) = \mathbf{l}_{n, m}(\bar{\mathbf{p}})} \end{aligned}$$

$\mathbf{L}(\mathbf{p}/\ell, \mathbf{e})$ is the position-dependent $6 \times 6 \times 3$ dimensionless spatial-distribution tensor with 18 unknown parameters, i.e., ℓ is an unknown constant and $\mathbf{e}$ is an unknown 17×1 vector. Let $\mathbf{L}_k$ represent the k -th symmetric 6×6 slice of the tensor $\mathbf{L}$ for $k \in \{1, 2, 3\}$,

$$\mathbf{L}(\bar{\mathbf{p}}; n, m, k) = \begin{bmatrix} \mathbf{L}_x(\bar{\mathbf{p}}; n, m) \\ \mathbf{L}_y(\bar{\mathbf{p}}; n, m) \\ \mathbf{L}_z(\bar{\mathbf{p}}; n, m) \end{bmatrix} = \begin{bmatrix} \mathbf{L}_1(\bar{\mathbf{p}}; n, m) \\ \mathbf{L}_2(\bar{\mathbf{p}}; n, m) \\ \mathbf{L}_3(\bar{\mathbf{p}}; n, m) \end{bmatrix} = \begin{bmatrix} \mathbf{L}_k(\bar{\mathbf{p}}; n, m) \\ k \in \{1, 2, 3\} \end{bmatrix}.$$

Current-based model:

$$f_k = \left(\frac{M g_B}{2\ell} \right) ({}^B g_I)^2 \mathbf{I}^T \bar{\mathbf{K}}^T \mathbf{L}_k(\mathbf{p}/\ell, \mathbf{e}) \bar{\mathbf{K}} \mathbf{I}$$

Calibration Approach:

1. Select N_p points in the work space.

$\mathbf{p}_i$ is the 3×1 position vector for for $i = 1, 2, \dots, N_p$.

$\mathbf{I}_j$ is the 6×1 input current vector for for $j = 1, 2, \dots, N_I$.

$\mathbf{v}_j$ is the 6×1 Hall sensor voltage vector for for $j = 1, 2, \dots, N_v$, wherein $N_v = N_I$.

$\mathbf{f}_{ij} = \mathbf{f}_j(\mathbf{p}_i)$ is the 3×1 resulting force vector for for $i = 1, 2, \dots, N_p$ and for $j = 1, 2, \dots, N_I$.

$$f_{ijk} = \left(\frac{M g_B}{2\ell} \right) ({}^B \hat{g}_I)^2 (\mathbf{I}_j^T \bar{\mathbf{K}}^T) \mathbf{L}_k(\mathbf{p}_i/\ell, \hat{\mathbf{e}}) (\bar{\mathbf{K}} \mathbf{I}_j) \text{ for } i = 1, 2, \dots, N_p \text{ and for } j = 1, 2, \dots, N_I,$$

wherein $^M g_B$ is the theoretical magnetization constant associated with the particle, and $\hat{\ell}$, $\hat{\mathbf{e}}$, $^B \hat{g}_I$, $\bar{\mathbf{K}}$, are previously obtained flux generation model parameters by calibration.

Voltage-based model:

$$f_k = {}^F g_V \mathbf{v}^T \bar{\mathbf{D}}^T \mathbf{L}_k(\mathbf{p}/\ell, \mathbf{e}) \bar{\mathbf{D}} \mathbf{v} = (\mathbf{v}_j^T {}^F \hat{\mathbf{H}}_V^T) \mathbf{L}_k(\mathbf{p}_i/\hat{\ell}, \hat{\mathbf{e}}) ({}^F \hat{\mathbf{H}}_V \mathbf{v}_j)$$

2. Cost function (Scenario one):

The k -th element of the vector $\boldsymbol{\varepsilon}_{ij}$ is:

$$f_{ijk} = \left(\frac{{}^M \hat{g}_B}{2\hat{\ell}} \right) ({}^B \hat{g}_I)^2 (\mathbf{I}_j^T \bar{\mathbf{K}}^T) \mathbf{L}_k(\mathbf{p}_i/\hat{\ell}, \hat{\mathbf{e}}) (\bar{\mathbf{K}} \mathbf{I}_j) \text{ (measurement)}$$

$$\varepsilon_{ijk}(\hat{\ell}, \hat{\mathbf{e}}, {}^F \hat{\mathbf{H}}_V) = f_{ijk} - (\mathbf{v}_j^T {}^F \hat{\mathbf{H}}_V^T) \mathbf{L}_k(\mathbf{p}_i/\hat{\ell}, \hat{\mathbf{e}}) ({}^F \hat{\mathbf{H}}_V \mathbf{v}_j)$$

$$\boldsymbol{\varepsilon}_{ij}^T \boldsymbol{\varepsilon}_{ij}(\hat{\ell}, \hat{\mathbf{e}}, {}^F \hat{\mathbf{H}}_V) = \sum_{k=1}^3 \left(f_{ijk} - (\mathbf{v}_j^T {}^F \hat{\mathbf{H}}_V^T) \mathbf{L}_k(\mathbf{p}_i/\hat{\ell}, \hat{\mathbf{e}}) ({}^F \hat{\mathbf{H}}_V \mathbf{v}_j) \right)^2$$

$$J(\hat{\ell}, \hat{\mathbf{e}}, {}^F \hat{\mathbf{H}}_V) = \sum_{j=1}^{N_I} \sum_{i=1}^{N_p} \boldsymbol{\varepsilon}_{ij}^T \boldsymbol{\varepsilon}_{ij} = \sum_{j=1}^{N_I} \sum_{i=1}^{N_p} \sum_{k=1}^3 \left(f_{ijk} - (\mathbf{v}_j^T {}^F \hat{\mathbf{H}}_V^T) \mathbf{L}_k(\mathbf{p}_i/\hat{\ell}, \hat{\mathbf{e}}) ({}^F \hat{\mathbf{H}}_V \mathbf{v}_j) \right)^2$$

$$\frac{\partial J}{\partial {}^F \hat{\mathbf{H}}_V} = -4 \sum_{j=1}^{N_I} \sum_{i=1}^{N_p} \sum_{k=1}^3 \left(f_{ijk} - (\mathbf{v}_j^T {}^F \hat{\mathbf{H}}_V^T) \mathbf{L}_k(\mathbf{p}_i/\hat{\ell}, \hat{\mathbf{e}}) ({}^F \hat{\mathbf{H}}_V \mathbf{v}_j) \right) \mathbf{L}_k(\mathbf{p}_i/\ell, \mathbf{e}) {}^F \hat{\mathbf{H}}_V \mathbf{v}_j \mathbf{v}_j^T$$

Let $\frac{\partial J}{\partial {}^F \hat{\mathbf{H}}_V} = 0$, can ${}^F \hat{\mathbf{H}}_V$ be expressed explicitly in terms of f_{ijk} , $\mathbf{L}_k$, and $\mathbf{v}_j$?

No, ${}^F \hat{\mathbf{H}}_V$ cannot be expressed explicitly in closed form.

3. Minimize cost function:

$$\text{Minimize } J(\hat{\ell}, \hat{\mathbf{e}}, {}^F \hat{\mathbf{H}}_V) \Rightarrow \hat{\ell}, \hat{\mathbf{e}}, {}^F \hat{\mathbf{H}}_V \text{ (Initial guess: } \hat{\ell} = \ell_{design}, \hat{\mathbf{e}} = \mathbf{0}, \text{ and } {}^F \hat{\mathbf{H}}_V = \sqrt{{}^M g_B / (2 \cdot \ell_{design})} {}^B \hat{g}_I \mathbf{I}_{6 \times 6}) \Rightarrow {}^F \hat{g}_V = \left(\frac{6}{5} h_{11} \right)^2 \text{ and } \bar{\mathbf{D}} = \frac{5}{6 \cdot h_{11}} {}^F \hat{\mathbf{H}}_V$$